

Leibnitz's Theorem

If each of the functions $u = u(x)$ and $v = v(x)$ are possessing n times derivatives, then

$$(uv)_n = u_n v + n_1 u_{n-1} v_1 + n_2 u_{n-2} v_2 + \dots + n_{n-1} u_1 v_{n-1} + u_0 v_n \quad (1)$$

where

$$u_n = \frac{d^n u}{dx^n}, \quad v_n = \frac{d^n v}{dx^n} \quad \& \quad (u_n)_n = \frac{d^n}{dx^n} (uv)$$

$$= D^n(u) \quad = D^n(v) \quad = D^n(uv)$$

"v" emon ekta ke dhoro jeta differentiate korle sesh hoye jabe

Example:

If $y = x^3 \sin x$ then find y_n .

Solⁿ:

Given, $y = x^3 \sin x$ ————— ①

By Leibnitz theorem applying D^n on both sides of ① we get

$$D^n y = D^n (x^3 \sin x)$$

$$\Rightarrow y_n = x^3 \sin\left(\frac{n\pi}{2} + x\right) + nC_1 (3x^2) \sin\left\{\frac{(n-1)\pi}{2} + x\right\} \\ + nC_2 (6x) \sin\left\{\frac{(n-2)\pi}{2} + x\right\} + nC_3 (6) \sin\left\{\frac{(n-3)\pi}{2}\right\}$$

$$= x^3 \sin\left(\frac{n\pi}{2} + x\right) - 3nx^2 \sin\left\{\frac{\pi}{2} - \left(\frac{n\pi}{2} + x\right)\right\} \\ - \frac{n(n-1)}{2} (6x) \sin\left\{\pi - \left(\frac{n\pi}{2} + x\right)\right\} \\ - \frac{n(n-1)(n-2)}{6} (6) \sin\left\{\frac{3\pi}{2} - \left(\frac{n\pi}{2} + x\right)\right\}$$

$$= x^3 \sin\left(\frac{n\pi}{2} + x\right) - 3nx^2 \cos\left(\frac{n\pi}{2} + x\right) - 3n(n-1)x \sin\left(\frac{n\pi}{2} + x\right) \\ + n(n-1)(n-2) \cos\left(\frac{n\pi}{2} + x\right)$$

$$= \left\{ x^3 - 3n(n-1)x \right\} \sin\left(\frac{n\pi}{2} + x\right) \\ - \left\{ 3nx^2 - n(n-1)(n-2) \right\} \cos\left(\frac{n\pi}{2} + x\right)$$

Ans

✓ Example:

If $y = x^3 \ln x$ then find y_n .

Sol: Given, $y = x^3 \ln x$ ——— ①

By Leibnitz theorem applying D^n on both sides of ① we obtain

$$D^n y = D^n (x^3 \ln x)$$

$$\Rightarrow y_n = x^3 \frac{(-1)^{n-1} (n-1)!}{x^n} + {}^nC_1 (3x^2) \frac{(-1)^{n-2} (n-2)!}{x^{n-1}}$$

$$+ {}^nC_2 (6x) \frac{(-1)^{n-3} (n-3)!}{x^{n-2}} + {}^nC_3 (6) \frac{(-1)^{n-4} (n-4)!}{x^{n-3}}$$

$$\Rightarrow y_n = \frac{x^3 (-1)^{n-1} (n-1)!}{x^n} + 3n x^2 \frac{(-1)^{n-2} (n-2)!}{x^{n-1}}$$

$$+ \frac{n(n-1)}{2} \frac{(6x) (-1)^{n-3} (n-3)!}{x^{n-2}} + \frac{n(n-1)(n-2)}{6} (6) \frac{(-1)^{n-4} (n-4)!}{x^{n-3}}$$

$$= \frac{(-1)^{n-1} (n-1)!}{x^{n-3}} + \frac{3n (-1)^{n-2} (n-2)!}{x^{n-3}} + \frac{3n(n-1) (-1)^{n-3} (n-3)!}{x^{n-3}}$$

$$+ \frac{n(n-1)(n-2)(-1)^{n-4}(n-4)!}{x^{n-3}}$$

$$\Rightarrow y_n = \frac{(-1)^{n-1}}{x^{n-2}} \left[(n-1)! - 3n(n-2)! + 3n(n-1)(n-3)! - n(n-1)(n-2)(n-4)! \right]$$

Ans

✓ Example:

If $y = \tan^{-1} x$, then show that

$$(i) (1+x^2) y_{n+1} + 2nxy_n + n(n-1)y_{n-1} = 0$$

$$(ii) (1+x^2) y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$$

Solⁿ:

$$(i) \text{ Given, } y = \tan^{-1} x \longrightarrow (1)$$

Diff. (1) wrt x we get

$$y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = 1 \longrightarrow (2)$$

By Leibnitz theorem applying D^n on both sides of (2)

$$D^n \{ (1+x^2) y_1 \} = D^n (1)$$

$$\Rightarrow (1+x^2) y_{n+1} + n \zeta (2x) y_n + n \zeta_2 (2) \cdot y_{n-1} = 0$$

$$\Rightarrow (1+x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} = 0$$

(Showed)

(ii) By Leibnitz theorem applying D^{n+1} on both sides of ② we obtain

$$D^{n+1} \{ (1+x^2) y_1 \} = D^{n+1} (1)$$

$$\Rightarrow (1+x^2) y_{n+2} + n+1 \zeta (2x) y_{n+1} + \frac{(n+1)(n+1-1)}{2} (2) y_n = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + 2(n+1) x y_{n+1} + n(n+1) y_n = 0$$

(Showed)

Example:

✓ If $y = \cot^{-1} x$, then show that

$$(i) (1+x^2) y_{n+1} + 2nx y_n + n(n-1) y_{n-1} = 0$$

$$(ii) (1+x^2) y_{n+2} + 2(n+1) x y_{n+1} + n(n+1) y_n = 0$$

Solⁿ:

(i) Given, $y = \cot^{-1} x$ ——— ①

Diff ① w.r.t x we obtain

$$y_1 = \frac{-1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = -1 \text{ ——— ②}$$

By Leibnitz theorem applying D^n on both sides of ② we get

$$D^n \{ (1+x^2) y_1 \} = D^n (-1)$$

$$\Rightarrow (1+x^2) y_{n+1} + n_1 C_1 (2x) y_n + n_2 C_2 (2) y_{n-1} = 0$$

$$\Rightarrow (1+x^2) y_{n+1} + 2nxy_n + n(n-1) y_{n-1} = 0$$

(Showed)

(ii) By Leibnitz theorem applying D^{n+1} on both sides of ② we obtain

$$D^{n+1} \{ (1+x^2) y_1 \} = D^{n+1} (-1)$$

$$\Rightarrow (1+x^2) y_{n+2} + n+1 C_1 (2x) y_{n+1} + n+1 C_2 (2) y_n = 0$$

$$\Rightarrow (1+x^2) y_{n+2} + 2(n+1)x y_{n+1} + n(n+1) y_n = 0$$

(Showed) ✓

Example:

✓ If $y = e^{a \sin^{-1} x}$, then show that

$$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

Solⁿ:

Given, $y = e^{a \sin^{-1} x}$ ——— ①

Diff ① wrt x we get

$$y_1 = e^{a \sin^{-1} x} \cdot \frac{a}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} y_1 = a e^{a \sin^{-1} x} = a y \quad [\text{using ①}]$$

$$\Rightarrow (1-x^2) y_1^2 = a^2 y^2 \quad \text{————— ②}$$

Diff ② again wrt x we get-

$$(1-x^2) 2y_1 y_2 - 2x y_1^2 = 2a^2 y y_1$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = a^2 y \quad \text{————— ③} \quad [\text{Dividing by } 2y_1]$$

By Leibnitz theorem applying D^n on both sides of ③ we have

$$D^n \{ (1-x^2) y_2 \} - D^n (x y_1) = D^n (a^2 y)$$

$$\Rightarrow (1-x^2) y_{n+2} + {}^nC_1 (-2x) y_{n+1} + {}^nC_2 (-2) y_n - x y_{n+1} - {}^nC_1 (1) y_n - a^2 y_n$$

$$\Rightarrow (1-x^2) y_{n+2} - 2nx y_{n+1} - n(n-1) y_n - 0x y_{n-1} - 0y_n - a^2 y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

(Showed)

✓ Example:

If $y = \sin(m \sin^{-1}x)$, then show that

$$(1-x^2) y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0.$$

Sol:

Given, $y = \sin(m \sin^{-1}x) \quad \text{--- (1)}$

Diff. (1) w.r.t x we get

$$y_1 = \cos(m \sin^{-1}x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} y_1 = m \cos(m \sin^{-1}x)$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 \cos^2(m \sin^{-1}x)$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 [1 - \sin^2(m \sin^{-1}x)]$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 (1-y^2) \quad \text{--- (2)}$$

Diff. ② again w.r.t x we obtain

$$(1-x^2) 2y_1 y_2 - 2x y_1^2 = m^2 (-2y_1 y_2)$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = -m^2 y_2 \quad [\text{Dividing by } 2y_1]$$

$$\Rightarrow (1-x^2) y_2 - x y_1 + m^2 y_2 = 0 \quad \text{--- (3)}$$

By Leibnitz theorem applying D^n on both sides of (3) we have

$$D^n \{(1-x^2) y_2\} - D^n (x y_1) + D^n (m^2 y_2) = D^n (0)$$

$$\Rightarrow (1-x^2) y_{n+2} + n_1 (-2x) y_{n+1} + n_2 (-2) y_n$$

$$- x y_{n+1} - n_1 (1) y_n + m^2 y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - 2n x y_{n+1} - n(n-1) y_n - x y_{n+1} - n y_n + m^2 y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1)x y_{n+1} + (m^2 - n^2) y_n = 0$$

(Showed) $\rightarrow$

Example: icourse'2024

If $\ln y = \tan^{-1} x$, then show that

$$(1+x^2) y_{n+2} + (2nx + 2x - 1) y_{n+1} + n(n+1) y_n = 0$$

Solⁿ:

Given, $\ln y = \tan^{-1} x$ ——— ①

Diff ① w.r.t x we get

$$\frac{1}{y} y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) y_1 = y \text{ ——— ②}$$

Again diff ② w.r.t x we obtain

$$(1+x^2) y_2 + 2xy_1 = y_1 \text{ ——— ③}$$

By Leibnitz theorem applying D^n on both sides of ③ we obtain

$$D^n \{ (1+x^2) y_2 \} + 2 D^n (xy_1) = D^n (y_1)$$

$$\Rightarrow (1+x^2) y_{n+2} + n_1 (2x) y_{n+1} + n_2 (2) y_n + 2xy_{n+1} + 2^n y_n = y_{n+1}$$
$$= y_{n+1}$$

$$\Rightarrow (1+x^2) y_{n+2} + 2nx y_{n+1} + n(n-1) y_n + 2xy_{n+1} + 2^n y_n = y_{n+1}$$

$$\Rightarrow (1+x^2) y_{n+2} + (2nx + 2x - 1) y_{n+1} + n(n+1) y_n = 0$$

(Showed) $\rightarrow$

✓ Example:

If $y = e^{a \cos^{-1} x}$, then show that

$$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + a^2) y_n = 0.$$

Soln:

Given, $y = e^{a \cos^{-1} x}$ ———— (1)

Diff (1) wrt x we obtain

$$y_1 = e^{a \cos^{-1} x} \cdot \left(\frac{-a}{\sqrt{1-x^2}} \right)$$

$$\Rightarrow \sqrt{1-x^2} y_1 = -a e^{a \cos^{-1} x} = -a y \quad [\text{using (1)}]$$

$$\Rightarrow (1-x^2) y_1^2 = a^2 y^2 \quad \text{———— (2)}$$

Again diff (2) wrt x we get

$$(1-x^2) 2y_1 y_2 - 2x y_1^2 = 2a^2 y y_1$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = a^2 y \quad [\text{Dividing by } 2y_1]$$

$$\Rightarrow (1-x^2) y_2 - x y_1 - a^2 y = 0 \quad \text{———— (3) ✓}$$

By Leibnitz theorem applying D^n on both sides of (3)

$$D^n \{ (1-x^2) y_2 \} - D^n (x y_1) - D^n (a^2 y) = 0$$

$$\Rightarrow (1-x^2) y_{n+2} + n_1 (-2x) y_{n+1} + n_2 (-2) y_n - x y_{n+1} - n_1 (1) y_n - a^2 y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - 2nx y_{n+1} - n(n-1)y_n - xy_{n+1} - xy_n - a^2 y_n = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2+a^2)y_n = 0$$

(Showed)

✓
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Example:

If $y = a \cos(\log x) + b \sin(\log x)$, then show that

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

Solⁿ:

Given, $y = a \cos(\log x) + b \sin(\log x)$ ——— ①

Diff ① w.r.t x we obtain

$$y_1 = -a \sin(\log x) \frac{1}{x} + b \cos(\log x) \frac{1}{x}$$

$$\Rightarrow xy_1 = -a \sin(\log x) + b \cos(\log x)$$
 ——— ②

Again diff ② w.r.t x we have

$$xy_2 + y_1 = -a \cos(\log x) \frac{1}{x} - b \sin(\log x) \frac{1}{x}$$

$$\Rightarrow x^2 y_2 + xy_1 = -[a \cos(\log x) + b \sin(\log x)]$$

$$\Rightarrow x^2 y_2 + xy_1 = -y \quad [\text{using ①}]$$

$$\Rightarrow x^2 y_2 + x y_1 + y = 0 \text{ --- (3)}$$

By Leibnitz rule applying D^n on both sides of (3)

$$D^n \{x^2 y_2\} + D^n \{x y_1\} + D^n \{y\} = D^n \{0\}$$

$$\Rightarrow x^2 y_{n+2} + {}^nC_1(2x) y_{n+1} + {}^nC_2(2) y_n + x y_{n+1} + {}^nC_1(1) y_n + y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + xy_{n+1} + ny_n + y_n = 0$$

$$\Rightarrow x^2 y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

(shown)

✓ Example:

Show that $\frac{d^{n+1}}{dx^{n+1}} (x^n \ln x) = \frac{n!}{x}$

Solⁿ:

Let $y = x^n \ln x$ --- (1)

Diff (1) w.r.t x we get

$$y_1 = nx^{n-1} \ln x + x^n \cdot \frac{1}{x}$$

$$\Rightarrow y_1 = \frac{nx^n \ln x}{x} + \frac{x^n}{x}$$

$$\Rightarrow xy_1 = nx^n \ln x + x^n$$

$$\Rightarrow xy_1 = nx + x^n \text{ [using (1)] --- (2)}$$

Now using Leibnitz theorem apply D^n on both sides of (2)

$$D^n \{xy\} = D^n(xy) + D^n(x^n)$$

$$\Rightarrow xy_{n+1} + n y(1) y_n = n y_n + n!$$

$$\Rightarrow xy_{n+1} + n y_n = n y_n + n!$$

$$\Rightarrow y_{n+1} = \frac{n!}{x}$$

$$\Rightarrow \frac{d^{n+1}}{dx^{n+1}}(y) = \frac{n!}{x}$$

$$\Rightarrow \frac{d^{n+1}}{dx^{n+1}}(x^n \ln x) = \frac{n!}{x}$$

(Showed)

Example:

If $y = (x^2 - 1)^n$, then prove that

$$(x^2 - 1) y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$$

Hence if $u = \frac{d^n}{dx^n} (x^2 - 1)^n$, then show that

$$(1 - x^2) \frac{d^2 u}{dx^2} - 2x \frac{du}{dx} + n(n+1)u = 0$$

Soln:

First Part: Let $y = (x^2 - 1)^n$ ——— (1)

Differentiating ① w.r.t x we get,

$$y_1 = n(x^2-1)^{n-1} \cdot 2x$$

$$\Rightarrow y_1 = \frac{n(x^2-1)^n \cdot 2x}{x^2-1}$$

$$\Rightarrow (x^2-1) y_1 = 2nx(x^2-1)^n$$

$$\Rightarrow (x^2-1) y_1 = 2nxy \text{ ——— ② [using ①]}$$

Again diff ② w.r.t x we obtain

$$(x^2-1) y_2 + 2xy_1 = 2nxy_1 + 2ny$$

$$\Rightarrow (x^2-1) y_2 + 2(1-n)xy_1 - 2ny = 0$$

Now, using Leibnitz theorem for n -th derivative, we get

$$D^n \{ (x^2-1) y_2 \} + D^n \{ 2(1-n)xy_1 \} - 2n D^n (y) = D^n (0)$$

$$\Rightarrow (x^2-1) y_{n+2} + {}^nC_1 (2x) y_{n+1} + {}^nC_2 (2) y_n$$

$$+ 2(1-n) [x y_{n+1} + {}^nC_1 (1) y_n] - 2n y_n = 0$$

$$\Rightarrow (x^2-1) y_{n+2} + 2nx y_{n+1} + n(n-1) y_n + 2(1-n)xy_{n+1} + 2(1-n)ny_n - 2ny_n = 0$$

$$\Rightarrow (x^2-1)y_{n+2} + 2xy_{n+1} + [n(n-1) + 2n(1-n) - 2n]y_n = 0$$

$$\Rightarrow (x^2-1)y_{n+2} + 2xy_{n+1} + [n^2-n + \cancel{2n-2n^2-2n}]y_n = 0$$

$$\Rightarrow (x^2-1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0 \quad \text{--- (3)}$$

(Proved)

2nd Part:

Equⁿ (3) can be rewritten as

$$(x^2-1) \frac{d^{n+2}}{dx^{n+2}}(y) + 2x \frac{d^{n+1}}{dx^{n+1}}(y) - n(n+1) \frac{d^n}{dx^n}(y) = 0$$

$$\Rightarrow \left((1-x^2) \frac{d^{n+2}}{dx^{n+2}}(y) - 2x \frac{d^{n+1}}{dx^{n+1}}(y) \right) + n(n+1) \frac{d^n}{dx^n}(y) = 0$$

$$\Rightarrow \frac{d}{dx} \left\{ (1-x^2) \frac{d^{n+1}}{dx^{n+1}}(y) \right\} + n(n+1) \frac{d^n}{dx^n}(y) = 0 \quad \text{--- (4)}$$

Now, $u = \frac{d^n}{dx^n}(x^2-1)^n = \frac{d^n}{dx^n}(y) \quad \text{[using (1)]}$

$$\therefore \frac{du}{dx} = \frac{d^{n+1}}{dx^{n+1}}(y)$$

Hence, equⁿ (4) becomes

$$\frac{d}{dx} \left\{ (1-x^2) \frac{du}{dx} \right\} + n(n+1)u = 0$$

$$\Rightarrow (1-x^2) \frac{d^2 u}{dx^2} - 2x \frac{du}{dx} + n(n+1)u = 0$$

(Proved)